

UNIT 1

Introduction

Sessions 1 and 2

Last year your machines answered yes or no. This year they will answer *how much*. That single change - from on-or-off to a whole range of values - is what turns a gadget into a real sensing machine.

By the end of this unit you will be able to:

- Explain the difference between a digital signal and an analog signal.
- Read a resistor's value from its colour bands.
- Recognise the VCC, GND, DO and AO pins that every sensor module uses.
- Name every component in your Class 7 kit and say what it is for.

SESSION 1

one class

The Digital World and the Analog World

Today you will

- Understand why some signals are on-or-off and others are a range.
- Learn where each idea is used in real machines.
- Meet the twelve components in your kit.

Think about a light switch on your classroom wall. It has exactly two positions: on and off. There is no halfway. Engineers call that a **digital** signal - it can only ever be one of two values, and there is nothing in between.

Now think about a fan regulator, or the brightness slider on a phone. These do not jump between two states. They move smoothly through hundreds of positions. Engineers call that an **analog** signal - a value anywhere within a range.

Here is why this matters. The real world is almost entirely analog. Temperature does not jump from cold to hot; it climbs. Light does not switch from day to night; it fades. Rain is not simply raining or not raining; there is drizzle, and there is a downpour. If your machine can only understand on-or-off, it can never describe the world properly.

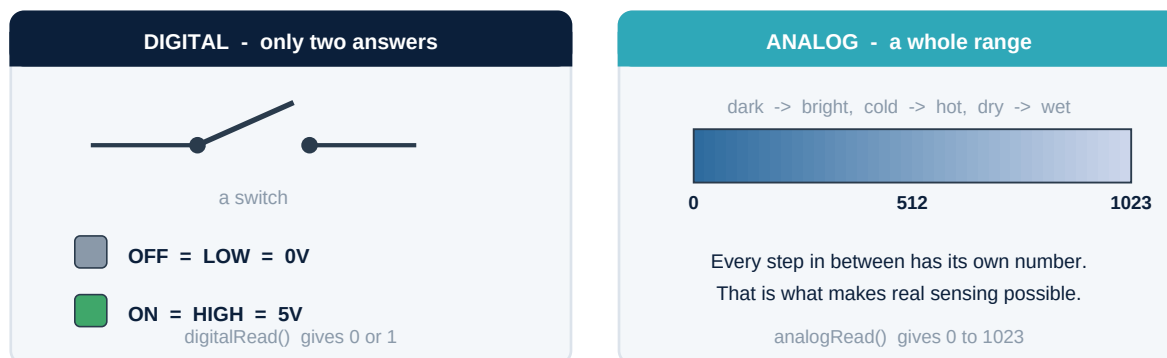


Figure 1.2 - The big idea of Class 7: numbers instead of yes-or-no

The Arduino handles both. Its digital pins, D0 to D13, deal in HIGH and LOW. Its analog pins, A0 to A5, measure a voltage and turn it into a number from 0 to 1023. Every project in this book depends on that second row of pins - which is exactly why Class 7 goes further than Class 6 did.

Try This

In your notebook, make two columns headed DIGITAL and ANALOG. Put these into the right column: a doorbell, a volume knob, a tubelight switch, room temperature, a keyboard key, the fuel gauge in a car, sunlight through a window. Be ready to defend any answer you are unsure about.

1.1 Open the Box

Lay everything out on the desk before you plug anything in. Learn the names first - it is impossible to ask a trainer for help with 'that small blue thing'. Notice that several of these parts are new this year: the potentiometer, the DHT11, the rain sensor and the servo motor.

ARDUINO KIT COMPONENTS – CLASS 7

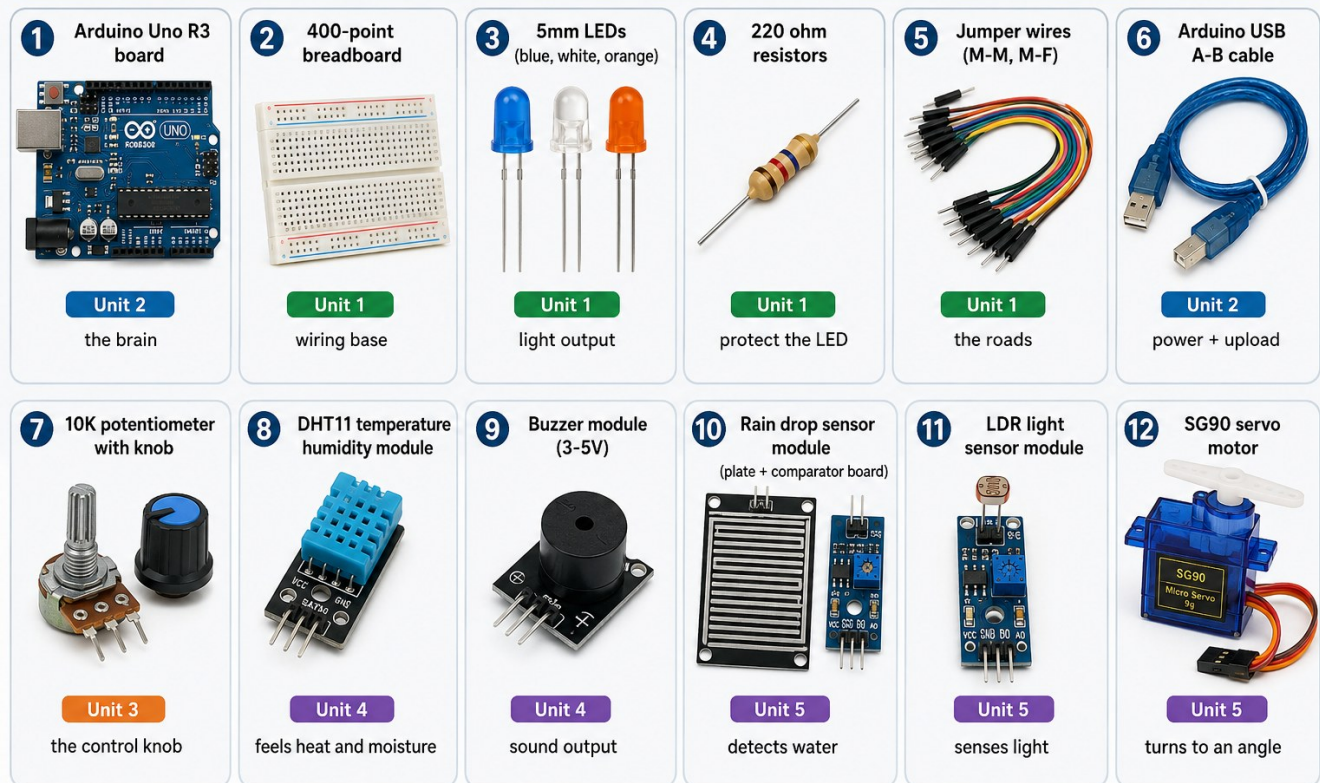


Figure 1.1 - Your Class 7 kit. Learn these twelve names before the next class.

Part	What it does	First used in
Arduino Uno board	The brain. It stores your program and runs it forever, even after the power has been off.	Session 3
USB A-B cable	Carries your program into the board and powers it while you test.	Session 3
400-point breadboard	Joins components together without any soldering.	Session 2
Jumper wires	The roads between parts. Male-to-male and male-to-female.	Session 2
5mm LEDs	Tiny lamps. They glow only one way round.	Session 2
220 ohm resistors	Limit the current so an LED is not destroyed.	Session 2
10K potentiometer	A knob that gives a smoothly changing value - your first analog input.	Session 6
DHT11 module	Measures temperature and humidity, both from one sensor.	Session 9
Buzzer module	Makes sound so the machine can warn people.	Session 11
Rain drop module	A plate that gets wet, plus a small board that reports how wet it is.	Session 12
LDR light sensor module	Reports how bright the surroundings are.	Session 13
SG90 servo motor	Turns to an exact angle and holds it. Used in the challenge work.	Session 15

Watch Out

The rain sensor comes in **two pieces**: the black plate with gold lines, and a small green control board. They join with two wires. Only the plate is allowed to get wet - never the control board, never the Arduino.

New word	What it means
Digital	A signal with only two possible values: on or off.
Analog	A signal that can take any value within a range.
Module	A small ready-made board with the fiddly parts already fitted.

SESSION 2

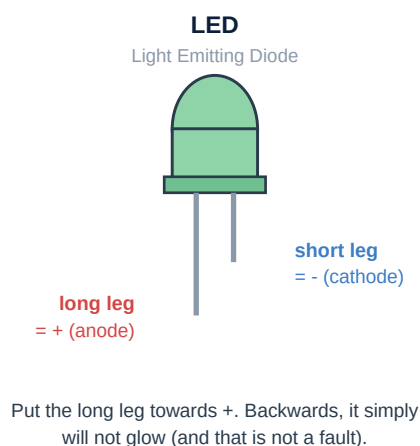
one class

Components, Resistor Colours and Modules

Today you will

- Read a resistor's value from its colour bands.
- Learn the four pins that almost every sensor module uses.
- Rebuild the LED circuit from Class 6 as a warm-up.

A resistor is a small part with a big job: it limits how much current can flow. Its value is measured in ohms, and it is printed on the body as coloured bands rather than numbers, because numbers would be too small to read. Each colour stands for a digit.



Its job is to protect the LED from too much current, like a narrow gate in a wide road.

Colour	Digit	Colour	Digit
Black	0	Green	5
Brown	1	Blue	6
Red	2	Violet	7
Orange	3	Grey	8
Yellow	4	White	9

The first two bands give you two digits. The third band tells you how many zeros to add after them. The fourth band, usually gold, is about accuracy and you can ignore it for now. So a resistor banded **red - red - brown** reads as 2, then 2, then one zero: **220 ohms**. A resistor banded **brown - black - orange** reads 1, then 0, then three zeros: **10,000 ohms**, which everybody calls 10K.

Try This

Find the 220 ohm and the 10K resistors in your kit without looking at the packet. Hold each one up and read the bands aloud to your partner. Then swap. This takes two minutes and saves you from a wrong resistor in every project this year.

2.1 The Four Pins Every Module Has

Look closely at the DHT11, the LDR module and the rain sensor board. They look completely different, but their pins follow the same pattern. Learn this pattern once and you can wire any sensor module you ever meet, including ones that do not exist yet.

Pin	Full name	What it is for	Connect it to
VCC	Voltage at the Common Collector	Power into the module	5V on the Arduino
GND	Ground	The return path for the electricity	GND on the Arduino
DO	Digital Out	A yes-or-no answer, set by the small blue screw on the board	any digital pin, D2 to D13
AO	Analog Out	A full reading from 0 to 1023	an analog pin, A0 to A5

Not every module has all four. The DHT11 has no AO pin at all, because it is a digital sensor that sends its readings as a coded pulse - you will meet that in Unit 4. But VCC and GND are always there, on every module, without exception.

Watch Out

The small blue box with a screw on many modules is a **trimmer**. Turning it changes the level at which the DO pin flips from LOW to HIGH. It has no effect at all on the AO pin. If you are using AO and someone tells you to 'adjust the screw', they are giving you the wrong advice.

Warm-up Activity - The Class 6 Circuit, From Memory

Before the coding starts again, prove your hands still remember. Build this without looking at any diagram.

- 1 Push an LED into the breadboard with its two legs in different columns.
- 2 Put a 220 ohm resistor in the same column as the long leg.
- 3 Wire the free end of the resistor to 5V, and the LED's short leg column to GND.
- 4 Plug in the USB. The LED should glow with no code at all.
- 5 Now move the resistor to the **short** leg side instead. Does it still work? Why?

Did You Know

Step 5 works perfectly. A resistor limits the current in the whole loop, so it does the same job wherever you place it in that loop. What matters is that it is *somewhere* in the path, not which side it sits on.

New word	What it means
Ohm	The unit of resistance.
Trimmer	A tiny adjustable screw that sets a module's DO threshold.
VCC	The power-in pin of a module. Goes to 5V.
AO	The analog-out pin. Gives a full reading, not just yes or no.